Advanced AI compact cheatsheet

STEP 1 Rational choice

1.1 Decision problems and preferences

Decision problem (A, X, c, ≽): actions, outcomes, consequence map c : A → X, preference. Utility goes on outcomes, v(a) = u(c(a)).

Three relations Weak x ≽ y. Strict x ≻ y means x ≽ y and not y ≽ x. Indifference x ∼ y means both directions.

Rationality Complete (every pair compared) plus transitive (rankings chain). Cycles have no best element and allow a money pump.

What carries over Transitivity of ≽ passes to ≻ and ∼. Completeness does not pass to either.

Indifference is not incompleteness Indifferent: equally good. Incomplete: cannot compare. Equality: the same object.

A decision problem $(A, X, c, \succeq)$

Weak preference $x \succeq y$

Strict preference
$x \succ y \iff x \succeq y \text{ and not } y \succeq x$

Indifference $x \sim y \iff x \succeq y \text{ and } y \succeq x$

Completeness
$\text{for all } x, y \in X: \quad x \succeq y \text{ or } y \succeq x$

Transitivity $x \succeq y \text{ and } y \succeq z \implies x \succeq z$

TRAP Equality and indifference are different ideas. Equality means two things are the same object. Indifference means two different outcomes are judged equally good. Tea and coffee are not equal, but you may be indifferent between them.

1.2 Utility functions

Representation x ≽ y ⟺ u(x) ≥ u(y). Strict gives >, indifference gives =. Not "=" for weak preference.

Ordinal Only order matters. Any strictly increasing f gives f(u), another valid utility. Gaps and ratios mean nothing.

Existence Finite X + complete + transitive ⟹ a utility exists. Build it by peeling off top classes: A ≻ B ∼ C ≻ D gives 3, 2, 2, 1.

Best exists In finite X an endless chain of improvements must cycle, which transitivity forbids. Ties at the top are allowed.

Counting n strict outcomes give n! rankings (5! = 120). With ties: 1, 3, 13, 75 weak orders.

Representation $x \succeq y \iff u(x) \geq u(y)$

Ordinal invariance
$\text{if } f \text{ is strictly increasing}, \quad v(x) = f(u(x))$
$\text{represents the same } \succeq$

Existence theorem
$X \text{ finite and } \succeq \text{ complete and transitive} \implies$
$\text{some } u : X \to \mathbb{R} \text{ represents } \succeq$

TRAP The slide transcript once read "u(x) = u(y) ⟺ x ≽ y". That is a transcription slip. The correct condition for weak preference uses ≽. Equality belongs to indifference only.

1.3 Choosing the best action

Rational choice v(a) = u(c(a)). a* ∈ arg max v, i.e. v(a*) ≥ v(a) for all a. Ties are all rational.

Movie tree Utilities 6 to 1, minus 1 per mile. INOX 1 mile, PVR 3. Best: INOX Aliens, net 5.

Fruit trees 1.6 Plain: AA at X − 60. Second same tree saves half: AA, OO become X + 5, PP X + 22.5, best AP at X − 55.

Parks 1.7 v = √(400c) − c/80 gives √c = 800, c* = 640,000 under the 1,000,000 cap. With √(1600c) the peak 2,560,000 is infeasible, so c* = 1,000,000.

Other ideas Satisficing picks the first v ≥ T. Minimizing loss = maximizing −loss. Check feasibility and boundaries.

Value of an action $v(a) = u(c(a))$

Rational choice
$a^* \in \arg\max_{a \in A} v(a) \iff v(a^*) \geq v(a)$
$\text{for every } a$

Spending after a plan
$u(\text{plan}) = X + \Sigma \text{ maintenance} - \Sigma \text{ reduction}$

Benefit $v(c) = \sqrt{400c} - \frac{c}{80} = 20\sqrt{c} - \frac{c}{80}$

First order condition $v'(c) = \frac{10}{\sqrt{c}} - \frac{1}{80} = 0$

TRAP The transcript sometimes wrote v(a*) > v(a) for every other a. That strict version only fits when the best action is unique. With ties the right condition is ≥.

STEP 2 Choice under risk

2.1 Lotteries and distributions

Lottery Action a gives $P(x_i \mid a) = p_i$ with p_i in [0, 1] and $\Sigma\, p_i = 1$. Nature picks the outcome.

CDF F(x) = P(X ≤ x | a). Non decreasing, 0 at −∞, 1 at +∞, right continuous. F(50) = 0.7 is "at most 50".

Intervals P(r < X ≤ s) = F(s) − F(r). Notes example 0.8 − 0.3 = 0.5.

Density F = ∫ f, f = F'. P(X = x) = 0 for continuous X. Densities can exceed 1.

Compound paths Multiply along each path, add the paths that reach the same outcome: 0.6 × 0.8 + 0.4 × 0.3 = 0.6.

Lottery of an action
$P(x_i \mid a) = p_i, \quad 0 \leq p_i \leq 1, \quad \sum p_i = 1$

CDF of an action $F_a(x) = P(X \leq x \mid a)$

Interval probability
$P(r < X \leq s \mid a) = F_a(s) - F_a(r)$

From density to CDF $F_a(x) = \int_{-\infty}^{x} f_a(t)\, dt$

TRAP F(50) = 0.7 means a 70% chance the outcome is at most 50. It does not mean a 70% chance the outcome is exactly 50. This misreading is the most common exam error on CDFs.

2.2 Expected utility and compound lotteries

Expected utility EU(a) = Σ P(x_i | a) u(x_i), or ∫ u dF. Choose arg max EU, not arg max E[X].

R&D compound lottery P(10 | G) = 0.625 × 0.9 + 0.375 × 0.5 = 0.75, EU(G) = 7.5 > EU(S) = 5. Multiply along paths, add paths.

Reversal u = 9, −1, 10, 0: EU(G) 6.5 > 5. u(B) = −8: 4.75 < 5. Same ranking, opposite choice.

Affine invariance Only v = a u + b with a > 0 keeps every EU comparison. Utility under risk is cardinal.

Other rules and PV Optimistic (P best), pessimistic (P worst). PV = FV ÷ $(1 + r)^t$. vNM axioms give EU.

Expected utility, discrete
$EU(a) = \sum_i P(x_i \mid a)\, u(x_i)$

Rational choice under risk
$a^* \in \arg\max_a \mathrm{E}[\, u(X) \mid a \,]$

Reduction of compound lotteries
$P(\text{final outcome}) = \sum_{\text{paths}} \prod_{\text{branches on the path}} P(\text{branch})$

Positive affine transformation
$v(x) = a\, u(x) + b, \quad a > 0$

EXAM Moving u(B) from −1 to −8 is not affine: it stretches one gap and leaves the others alone. Under risk the sizes of the gaps matter, so expected utility needs cardinal utility, unique up to a u + b with a > 0.

2.3 Risk attitudes and the St Petersburg paradox

Three attitudes Concave u (ln, √): averse, E[u] ≤ u(E[X]). Linear: neutral. Convex (x^2): loving. Jensen with the chord picture.

Certainty equivalent u(CE) = E[u(X)]. Risk premium = E[X] − CE. √x with 100 at 0.2: EU = 2, CE = 4, sure 30 wins.

Fair gamble +1 or −1 at 50/50 has mean 0. Play when 0.5u(1) + 0.5u(−1) > u(0). Averse agents refuse.

St Petersburg Pays 2^{n-1} w.p. $1/2^n$. E[X] = Σ ½ = ∞. Log utility: E = ln 2, CE = 2. x^2 still diverges.

Practice 5 u = x^2, 0 or 10: E[X] = 5, u(E[X]) = 25, E[u] = 50, so risk loving.

Risk neutral $\mathrm{E}[\, u(X) \,] = u(\, \mathrm{E}[X] \,)$

Risk averse $\mathrm{E}[\, u(X) \,] \leq u(\, \mathrm{E}[X] \,)$

Risk loving $\mathrm{E}[\, u(X) \,] \geq u(\, \mathrm{E}[X] \,)$

Certainty equivalent
$u(\mathrm{CE}) = \mathrm{E}[\, u(X) \,] \implies \mathrm{CE} = u^{-1}(\, \mathrm{E}[u(X)] \,)$

Expected money
$\mathrm{E}[X] = \sum_{n=1}^{\infty} \frac{1}{2^n} \cdot 2^{n-1} = \sum \frac{1}{2} = \infty$

Expected log utility
$\mathrm{E}[\ln X] = \sum \frac{(n-1)\ln 2}{2^n} = \ln 2$

EXAM Concavity is what tames the paradox. With the convex u = x^2 the terms are 2^{n-2} and the sum still diverges. Even √x converges, to about 1.707, which gives a certainty equivalent near 2.91.

STEP 3 Decisions over time

3.1 Backward induction

Method From the last decision back: chance node = Σ p × value, decision node = max. Never compare raw leaves at a chance node.

R&D tree Success: m 11 > d 8. Failure: m 3 < d 4. Go = 0.625 × 11 + 0.375 × 4 = 8.375 > skip 5.

Full strategies MM 8.0, MD 8.375, DM 6.125, DD 6.5. Same answer as folding, but 2^k plans for k binary nodes.

Practice A Good buzz push 14.4, weak buzz hold 3.7, big budget 7.98 > small 3.5.

Why it works Bellman's principle of optimality: each tail of an optimal plan is optimal. Folding is linear in the tree size.

Backward induction
start at the last decision, replace each chance node by its expected value, keep the best action at each decision node, and move one level back

TRAP Never compare raw leaf payoffs where nature moves. 13 is not "better than" −7 in any useful sense at a chance node, because the firm does not choose between them. Convert every chance node to its expected value first, and compare only at nodes where the firm chooses.

3.2 Discounting and consumption over time

Discount factor δ = 1/(1 + r), 0 < δ < 1, patience not a choice. A after n periods is worth Aδⁿ. Practice B: PV = 520.2.

Timing Discount each piece by when it happens: R&D branch 10δ − 1, not (10 − 1)δ.

ln consumption 2 periods K/(1 + δ), δK/(1 + δ). 3 periods K, δK, δ²K over 1 + δ + δ². Equal discounted marginal utility.

Corners Linear u: (K, 0, 0). Convex u = x²: FOC finds the minimum, best is a corner. √x: $x_1 = K/(1+\delta^2)$, K = 181 gives 100, 81.

Time consistency Exponential: re-planning keeps x_2^*, x_3^*. Hyperbolic 1/(1 + kt) is time inconsistent.

Discount factor $\delta = \dfrac{1}{1+r}, \quad 0 < \delta < 1$

n periods ahead
value today of A received after n periods $= A\,\delta^n = \dfrac{A}{(1+r)^n}$

Two periods, u = ln x
$\max \ln x_1 + \delta \ln(K - x_1) \implies x_1^* = \dfrac{K}{1+\delta}, \quad x_2^* = \dfrac{\delta K}{1+\delta}$

Three periods, u = ln x
$x_1^* = \dfrac{K}{1+\delta+\delta^2}, \quad x_2^* = \dfrac{\delta K}{1+\delta+\delta^2}, \quad x_3^* = \dfrac{\delta^2 K}{1+\delta+\delta^2}$

3.3 The value of information

VOI formula VOI = E[max_a u(a, s)] − max_a E[u(a, s)]. Choose per state then average, minus average per action then choose.

MBA Cost 10, states 0.25/0.5/0.25. No info: 9 (MBA) vs 8. Oracle: 22, 8, 4 gives 10.5. VOI = 1.5.

Never negative You can ignore the oracle, so VOI ≥ 0. It is 0 when one action is best in every state.

Venue practice C Outdoor 50/−20, indoor 30/25 at 0.5 each. 27.5 without, 37.5 with, VOI = 10. 90% forecast: 6.75.

Imperfect signals Update by Bayes, pick the best per signal, average. Worth 0 if it cannot change the decision (q = 0.5).

Value of perfect information
$\mathrm{VOI} = \mathrm{E}[\,\max_a u(a, s)\,] - \max_a \mathrm{E}[\,u(a, s)\,]$

MBA example VOI = 10.5 − 9 = 1.5

Bayes update
$P(s \mid \text{signal}) = \dfrac{P(\text{signal} \mid s)\,P(s)}{P(\text{signal})}$

TRAP A student in class computed 0.25 × 22 + 0.5 × 8 + 0.25 × 4 as the no information value. That mixes actions across states, which you can only do once you know the state. Without information you must use one action in every state.

STEP 4

Games between rational agents

4.1 Static games and the normal form

Two axes Complete vs incomplete information. Static (no one sees others' choices) vs dynamic. Static is about observation, not clock time.

Common knowledge Everyone knows, everyone knows everyone knows, forever. Complete info: actions, outcomes, map, preferences are common knowledge.

Normal form ⟨N, S_i, v_i⟩ with v_i on full profiles. Finite if all S_i finite. Bimatrix cell = (row payoff, column payoff).

Prisoner's Dilemma (M,M) = (−2,−2), (M,C) = (−5,−1), (C,M) = (−1,−5), (C,C) = (−4,−4). C is the best reply to both.

Best response $\mathrm{BR}_i(s_{-i})$ maximizes $v_i(\cdot, s_{-i})$. A profile with no profitable unilateral deviation is self enforcing.

Common knowledge
E is common knowledge $\iff$ (everyone knows)^n E for every $n = 1, 2, 3, \ldots$

Normal form game $\langle\, N, \{\{S_i\}\}, \{\{v_i\}\}\,\rangle$

Payoff function $v_i : S_1 \times \ldots \times S_n \to \mathbb{R}$

Best response
$s_i \in \mathrm{BR}_i(s_{-i}) \iff v_i(s_i, s_{-i}) \geq v_i(s'_i, s_{-i})$ for all s'_i

4.2 Modelling games: Cournot, Bertrand and the streetlamp

Recipe Players, strategies (with their range), payoffs split into cases. Game = ⟨N, S_i, v_i⟩.

Cournot P = a − bq. s_i = (a − c_i − bs_j)/2b, s* = (a − c)/3b. a = 100, c = 10: 30 each, P = 40, profit 900.

Asymmetric Cournot s_1^* = (a − 2c_1 + c_2)/3b. Practice F: 18 and 12, P = 30, profits 324 and 144.

Bertrand S_i = [0, a], q = (a − p)/b. Cheaper takes all, tie splits (divide by 2b), dearer gets $-C_i(0)$. Paradox: p = c.

Streetlamp $v_i = 3 \cdot 1[\Sigma \geq 2] - s_i$, 8 profiles. Pay only when pivotal. Stable: exactly two payers, or nobody.

Cournot payoff
$v_i(s_i, s_j) = (a - b(s_i + s_j))\,s_i - C_i(s_i)$, or $-C_i(s_i)$ if the price would be negative

Best response, linear cost c_i
$s_i = \dfrac{a - c_i - b\,s_j}{2b}$

Symmetric equilibrium $s^* = \dfrac{a - c}{3b}$

Bertrand payoff
$v_i = p_i \dfrac{a - p_i}{b} - C_i\!\left(\dfrac{a - p_i}{b}\right)$ if cheaper, half the market (divide by $2b$) if tied, $-C_i(0)$ if dearer

Streetlamp payoff
$v_i(s) = 3 \cdot 1[\,\sum_j s_j \geq 2\,] - s_i, \quad s_i \in \{\{0, 1\}\}$

EXAM The Bertrand paradox: with equal unit cost c, the only stable prices are $p_1 = p_2 = c$. Two firms are enough to drive profit to zero, while Cournot firms keep the price at 40, well above cost 10.

4.3 Solution concepts and Pareto optimality

Solution concept Narrows profiles to reasonable ones. Its pick is an equilibrium. Not a physics equilibrium.

Four assumptions Rational (maximize), intelligent (know the game), common knowledge of both, self enforcement (no outside enforcer).

Good concept Existence, uniqueness, invariance. "Anything can happen" exists but is not unique.

Pareto s dominates s': ≥ for all, > for one. PD: only (C, C) fails. Marketing War: (L,L), (L,M), (M,L). Zero sum: all optimal.

Altruism $v_i + \tfrac{1}{2}v_j$ gives (−3,−3), (−5.5,−3.5), (−3.5,−5.5), (−6,−6). Mum becomes best. Pareto is not fairness.

Pareto dominance
s Pareto dominates $s' \iff v_i(s) \geq v_i(s')$ for all i, and > for at least one j

Pareto optimal
s' is not Pareto dominated by any profile

Altruistic payoff $v_i^{\text{alt}}(s) = v_i(s) + \tfrac{1}{2}\,v_j(s)$

TRAP Pareto optimality says nothing about fairness. (2, 8) and (8, 2) are Pareto optimal next to the balanced (6, 6). It only rules out waste.

STEP 5

Dominance and iterated elimination

5.1 Dominance and the dominant strategy equilibrium

Strict domination $s_i \rhd s'_i$: $v_i(s_i, s_{-i}) > v_i(s'_i, s_{-i})$ for every s_{-i}. Row player: first numbers across columns. Column player: second numbers down rows.

Vocabulary Dominating/dominated: beats/is beaten by at least one. Dominant/submissive: beats/is beaten by all. One dominator is enough to delete.

Partial order ▷ is irreflexive, asymmetric, transitive, not complete (RPS, M and H in the Marketing War). Weak: ≥ (textbook adds > once).

DSE Everyone plays a strictly dominant strategy. Unique if it exists. PD (C, C), altruistic PD (M, M), RPS and Marketing War none.

Rationality A rational player never plays a strictly dominated strategy under any belief. Increasing transforms and $h(s_{-i})$ do not change dominance.

Strict domination
$s_i \rhd_i s'_i \iff v_i(s_i, s_{-i}) > v_i(s'_i, s_{-i})$ for every s_{-i}

Strict dominant strategy equilibrium
s^D with s_i^D strictly dominant for every player i

Rational players and dominated strategies
$E_\mu[v_i(s_i)] - E_\mu[v_i(s'_i)] = \sum \mu(s_{-i})[v_i(s_i, s_{-i}) - v_i(s'_i, s_{-i})] > 0$

TRAP The mistake made twice in class: checking player 1 by mixing in the second number of each cell, or by moving down a column. Rule: for the row player compare first numbers, moving across columns. For the column player compare second numbers, moving down rows. Moving along the other player's strategies is exactly "for every s_{-i}".

5.2 Iterated elimination of dominated strategies

Algorithm Each round delete every strictly dominated strategy of every player, comparing only survivors. Stop when nothing is dominated.

Common knowledge Round k uses k layers of "I know that you know". Real people manage about 1 to 3 layers.

Slide 11 R ▷ C, then U ▷ M and D, then L ▷ R. Outcome (U, L) = (4, 3), self enforcing but Pareto dominated by (9, 6).

Theorems DSE found in one round. Every NE survives. Unique survivor = unique NE. Strict IESDS is order independent.

Limits Marketing War leaves {M, H}², RPS deletes nothing. Weak IESDS depends on order: (2, 1) or (1, 1).

IESDS, round k
$DS_i^k = \{\{ s_i \in S_i^k : \text{some } s'_i \in S_i^k \text{ beats it against every } s_{-i} \in S_{-i}^k \}\}$

Update $S_i^{k+1} = S_i^k \setminus DS_i^k$

Nash equilibrium (preview)
$v_i(s_i^*, s^*_{-i}) \geq v_i(s_i, s^*_{-i})$ for every i and every s_i

5.3 Dominance at work: Cournot, auctions and voting

Cournot IESDS BR = max{0, (a − c − bq)/2b}. Rounds: [0, 45], [22.5, 45], [22.5, 33.75], ... halving to (a − c)/3b = 30. E1: 40, price 50, profit 1,600.

Second price auction Payoff v − B if b > B. Bidding v weakly dominates every bid (never worse, sometimes better). Strategy proof.

First price auction Payoff v − b if you win. Overbidding weakly dominated, but bidding v not dominant: bidders shade.

Location and prices Hotelling 1 to 5: delete 1, 5 then 2, 4, both at 3 (median). Discrete Bertrand: strict deletes nothing, weak ends at (2, 2) losing (1, 1), (3, 3).

Weak only Below cost Bertrand prices and the guess game: weakly but not strictly dominated, so strict IESDS cannot start.

Best response
$\mathrm{BR}_i(q_j) = \max\{\{ 0, \frac{a - c - bq_j}{2b} \}\}$

Second price (Vickrey)
payoff = $v - B$ if $b > B$, otherwise 0

First price
payoff = $v - b$ if $b > B$, otherwise 0

STEP 6 Quiz and exam revision

6.1 Quiz 1 solved: all four questions

First order condition $\frac{a(q_j + 1)}{(q_i + q_j + 1)^2} = c$

Symmetric equilibrium
$q^* = \frac{(a - 4c) + \sqrt{a^2 + 8ac}}{8c}$

Joint monopoly $Q^* = \sqrt{\frac{a}{2c}} - 1$

Buy one more unit when
marginal value ≥ price

Buy when
$0.4\sqrt{400 - p + 900} + 0.6\sqrt{400 - p + 10} \geq \sqrt{400} = 20$

Break even price $p^* \approx 267.99$

Answer $(x_1, x_2, x_3) = (K, 0, 0), \quad v = K^2$

EXAM The trap in S1 is pattern matching. (a − c)/3 is the textbook Cournot answer, but it comes from demand a − bq. With a different demand curve you must redo the first order condition.

TRAP This question has two true options under the standard reading (prizes added to wealth after paying p). If only one option may be marked, A is the safer choice because it covers every price below 18, and B is a special case. It is worth asking the instructor whether the question allows more than one answer.

6.2 Exam revision: from school to research universities

Expected value $E[X] = \Sigma\, p_i\, x_i$

Present value $PV = \Sigma\, \delta^t\, x_t$

EXAM Round only at the very end. A numerical answer type key often has a narrow range such as 52.6 to 52.7.

EXAM In proofs, name each assumption when you use it: completeness, transitivity, the vNM axioms, common knowledge of rationality. Examiners often ask what breaks when one fails.